MODULE 43 · EXAM STRATEGY - ALL SECTIONS

Practice Paper 1

Twenty new questions across all five sections, in exam proportions, each followed by a worked answer. Pause, choose, then check.

SECTIONS All five MODULE 43 of 45 QUESTIONS 20

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How to use this paper

- Each question pauses for **nine seconds**. Pause the video if you need longer.
- The real exam allows about **2.5 minutes** per question. Practise reading the ask first.
- Write down your **letter** before each answer is revealed.
- Sections are **mixed**, as on the exam. The score slide maps each question to its section.

> REMEMBER
>
> Score yourself honestly, every time. A lucky guess you cannot explain counts as a miss.

> The readings pipeline sums kWh in 30-minute event-time windows. After an outage, some meters send readings up to two hours late.
>
> Late readings must still update the totals, without holding the first results back.

What should the team configure?

A Processing-time windows, so late readings land in the current window

B Allowed lateness of two hours, with a late trigger per late reading

C A global window with a single trigger once a day at midnight

D Drop late readings, and correct the totals in a nightly batch job

◆ PAUSE AND DECIDE

What should the team configure?

A Processing time

Readings land in the wrong half hour.

B Allowed lateness

On-time result first, then updates as late data lands.

C Global window

Every result held back for a day.

D Nightly correction

Two pipelines, and totals wrong until night.

THE SIGNAL

Late data must still count, without delay: allowed lateness plus late firings.

> Larkspur is creating new BigQuery datasets for German customer data. The data protection agreement says the data must be stored only in Germany.
>
> The team also wants to stop anyone creating German resources elsewhere by mistake.

What should the team do?

A Create the datasets in the EU multi-region, which covers Germany

B Create them in europe-west3, and rely on a naming convention

C Use VPC Service Controls to keep the data inside Germany

D europe-west3, plus a resource locations policy on the de folder

◆ PAUSE AND DECIDE

What should the team do?

A EU multi-region

Spans many countries, not just Germany.

B Naming convention

Conventions do not stop mistakes.

C VPC Service Controls

Controls movement, not where data is stored.

D Location plus policy

Frankfurt, and new resources blocked elsewhere.

THE SIGNAL

Where data may be created: dataset location plus the resource locations constraint.

> The live readings table in Bigtable is queried for one meter's readings over the last day.
>
> The first design used the reading timestamp as the row key, and one node now runs hot.

Which row key should replace it?

A meter_id, followed by a reversed timestamp

B The reading timestamp, followed by meter_id

C A hash of meter_id and timestamp, for an even spread

D A sequential number assigned to each reading

◆ PAUSE AND DECIDE

Which row key should replace it?

A **Meter, reversed time**
Writes spread, and each meter's newest rows sit together.

B **Timestamp first**
Still leads with time, so still hot.

C **Hashed key**
Even spread, but per-meter range scans are lost.

D **Sequential number**
Every new write lands on one node.

THE SIGNAL

Queries by meter over time: meter first, then a reversed timestamp.

> Regional managers query one BigQuery table of bills. Each manager must see only the rows for their own region.
>
> The table must not be copied, and managers use their own SQL as well as dashboards.

What should the team set up?

A One table per region, copied each night from the main table

B Policy tags on the region column, with Fine-Grained Reader per group

C Row access policies on region, one per manager group

D A dashboard filter in Data Studio, set to each manager's region

◆ PAUSE AND DECIDE

What should the team set up?

A **Table per region**
Copies, which the brief forbids.

B **Policy tags**
Column security, not row filtering.

C **Row access policies**
Each group sees only its own region's rows.

D **Dashboard filter**
Their own SQL sees everything.

THE SIGNAL

Same table, different rows per person: row-level security.

> Westgrid must receive regional demand within five minutes of each half hour. The readings pipeline runs on Dataflow.
>
> Operations want to be paged as soon as the pipeline's output starts falling behind event time.

What should they alert on?

A The job's data freshness, above five minutes

B Worker CPU utilisation staying above eighty percent

C A daily scheduled query counting the late rows

D The number of workers reaching the autoscaling maximum

◆ PAUSE AND DECIDE

What should they alert on?

A **Data freshness**
Measures exactly how far output trails event time.

B **Worker CPU**
High CPU can be healthy. Low CPU can still be stuck.

C **Daily query**
A day too late to page anyone.

D **Worker count**
Capacity, not staleness.

THE SIGNAL

Output falling behind event time: alert on data freshness.

> Security worries that a stolen analyst credential could be used from outside the company to copy BigQuery data into a personal project.
>
> IAM roles are already least privilege.

What adds the missing protection?

A Remove BigQuery Data Viewer from every analyst in the company

B A VPC Service Controls perimeter around the data projects

C Customer-managed encryption keys on every BigQuery dataset

D Rotate every analyst's password once a week

◆ PAUSE AND DECIDE

What adds the missing protection?

A Remove access

Stops the work, not the theft.

B Perimeter

Copying out is blocked, even with valid credentials.

C CMEK

Authorised reads still decrypt the data.

D Password rotation

A stolen token works until it expires.

THE SIGNAL

Exfiltration with valid credentials: a VPC Service Controls perimeter.

A billing subscriber processes meter events from Pub/Sub. Events for the same meter must be processed in the order they were published.

Different meters can be processed in any order, and the subscriber must scale out.

What should the team do?

A Add a sequence number, and sort each batch in the subscriber

B Use a single subscriber with one thread, so nothing runs in parallel

C Create a separate topic for every meter

D Publish with the meter ID as the ordering key, and enable ordering

◆ PAUSE AND DECIDE

What should the team do?

A Sort each batch

Batches can split one meter's events.

B One thread

Ordered, but it cannot scale.

C Topic per meter

Millions of topics to manage.

D Ordering key

Per-meter order, parallel across meters.

THE SIGNAL

Order per entity, scale across entities: an ordering key.

> Settlement files in Cloud Storage are read often for 30 days, then about once a quarter for audits.
>
> They must be deleted after seven years. The team wants the lowest cost with no manual work.

What should the team set up?

A Keep everything in Standard, and delete old files by hand each year

B Archive storage from day one, with a seven-year retention policy

C Lifecycle rules: Coldline at 30 days, delete at seven years

D Nearline from day one, with object versioning turned on

◆ PAUSE AND DECIDE

What should the team set up?

A All Standard

Pays Standard prices for years of cold data.

B Archive at once

Retrieval fees all through the busy first month.

C Lifecycle rules

Cheap for quarterly reads, deleted automatically.

D Nearline and versions

Versions add cost, and nothing ages out.

THE SIGNAL

Access falls with age: lifecycle rules. Quarterly reads: Coldline.

> The customer accounts database runs on Cloud SQL for MySQL. Analysts want its inserts, updates and deletes in BigQuery within minutes.
>
> The team wants no custom code to maintain.

What should the team use?

A. A nightly export to Cloud Storage, loaded into BigQuery

B. Federated queries from BigQuery against Cloud SQL

C. Database Migration Service, replicating into BigQuery

D. Datastream, with BigQuery as the destination

◆ PAUSE AND DECIDE

What should the team use?

A Nightly export

A day stale.

B Federated queries

Live, but every query loads production.

C DMS

Migrates to Cloud SQL or AlloyDB, not BigQuery.

D Datastream

Change data capture, merged in within minutes.

THE SIGNAL

Changes into BigQuery continuously, with no code: Datastream.

A new Dataflow job reads one Pub/Sub subscription and writes one BigQuery dataset.

Its workers currently run as the Compute Engine default service account.

What is the Google-recommended identity for its workers?

A Its own service account: Dataflow Worker plus source and sink roles

B The Compute Engine default service account, with Editor on the project

C The personal user account of the developer who wrote the pipeline

D A service account key file stored in the pipeline's Git repository

◆ PAUSE AND DECIDE

What is the Google-recommended identity for its workers?

A Dedicated account
Least privilege, one identity per job.

B Default with Editor
Far too broad, shared by everything.

C Personal account
Grants their access, and breaks when they leave.

D Key in Git
A leaked secret waiting to happen.

THE SIGNAL

Google-recommended worker identity: dedicated and least privilege.

> Analysts want a daily demand forecast for each region, 30 days ahead, from three years of history already in BigQuery.
>
> They know SQL, but not Python.

What should they use?

A Export the history, and train a model in a Python notebook first

B A linear regression model on the date column, in BigQuery ML

C An ARIMA_PLUS model in BigQuery ML, with ML.FORECAST per region

D A k-means model in BigQuery ML, clustering days by demand

◆ PAUSE AND DECIDE

What should they use?

A Python notebook

Not SQL, and the data leaves BigQuery.

B Linear regression

Ignores seasonality and trend.

C ARIMA_PLUS

A time series per region, in SQL, with seasonality.

D k-means

Clustering, not forecasting.

THE SIGNAL

Forecast a time series in SQL: ARIMA_PLUS with ML.FORECAST.

> Larkspur's analysts use on-demand pricing. One analyst's mistaken query loop scanned 300 TB in a single day.
>
> Finance wants a hard daily cap for each user, without slowing anyone else down.

What should the team set?

A. A budget alert at 80 percent of the monthly BigQuery spend

B. A custom quota on query usage per user per day

C. Move the whole project to an Enterprise edition reservation

D. Require a partition filter on every table in the project

◆ PAUSE AND DECIDE

What should the team set?

A Budget alert

Alerts, but stops nothing.

B Per-user quota

A daily limit per user, enforced.

C Reservation

Changes billing to cap one mistake.

D Partition filter

Helps some queries, caps nothing.

THE SIGNAL

A hard cap per user per day, on-demand: a custom quota.

> The raw readings table is partitioned by reading date. Regulations require only 400 days of raw data.
>
> Older partitions should disappear automatically, with no code to run.

What should the team configure?

A. A scheduled query that deletes rows older than 400 days each night

B. A table expiration of 400 days on the raw readings table

C. A partition expiration of 400 days on the table

D. Time travel set to its maximum, so old data ages out

◆ PAUSE AND DECIDE

What should the team configure?

A Nightly delete

Code to maintain, and DML to pay for.

B Table expiration

Deletes the whole table at 400 days.

C Partition expiration

Each day's partition drops at 400 days.

D Time travel

Recovery, seven days at most.

THE SIGNAL

Drop old data by age, automatically: partition expiration.

> Larkspur must move 400 TB of archived settlement history from its data centre to Cloud Storage.
>
> The only link is 200 Mbps, already half used. The move must finish within two months.

How should the data move?

A gcloud storage cp, running in parallel over the existing link

B Storage Transfer Service agents, scheduled to run overnight

C A new VPN tunnel, dedicated to the transfer

D Transfer Appliance, shipped and loaded at the data centre

◆ PAUSE AND DECIDE

How should the data move?

A gcloud storage

About a terabyte a day: over a year.

B Transfer agents

The same thin link, the same year.

C VPN tunnel

A tunnel adds no bandwidth.

D Transfer Appliance

Weeks, whatever the link.

THE SIGNAL

Hundreds of terabytes over a thin link: do the arithmetic, then ship it.

> Larkspur's on-premises Hadoop cluster runs 300 Spark settlement jobs that read from HDFS.
>
> The team wants them running on Google Cloud quickly, with minimal code changes.

What should the team do first?

A Rewrite every job as an Apache Beam pipeline on Dataflow

B Run the jobs on Managed Spark, with the data in Cloud Storage

C Convert every job into BigQuery SQL, scheduled by Dataform

D Rebuild the same Hadoop cluster on Compute Engine virtual machines

◆ PAUSE AND DECIDE

What should the team do first?

A Rewrite in Beam

Three hundred rewrites.

B Managed Spark

Same Spark code, Cloud Storage paths instead of HDFS.

C Rewrite in SQL

Another three hundred rewrites.

D Hadoop on VMs

Moved, but the team still runs the cluster.

THE SIGNAL

Existing Spark, quickly, few changes: Managed Spark with Cloud Storage.

> Westgrid, a separate company, wants daily read access to Larkspur's regional demand mart. Larkspur wants no copies of the data, and Westgrid should pay for its own queries.

How should Larkspur share it?

A. Publish the mart as a listing in a BigQuery sharing exchange

B. Export the mart to Cloud Storage each night, and share the bucket

C. Grant Westgrid's users Data Viewer on Larkspur's whole project

D. Copy the mart into a project that Westgrid owns, every night

◆ PAUSE AND DECIDE

How should Larkspur share it?

A Sharing listing
Linked, read-only, current, and the subscriber pays.

B Nightly export
A copy, and stale by morning.

C Project-wide access
Far too broad, across the whole project.

D Nightly copy
A copy, which Larkspur wants to avoid.

THE SIGNAL

Another company, no copies, they pay: a sharing listing.

> Larkspur has hundreds of BigQuery tables built up over years. The privacy team suspects some hold unlabelled personal data, such as phone numbers and bank details.
>
> They want to find it with the least effort, and keep finding it as tables change.

What should they do?

A Ask every data owner to review all of their tables by hand

B Write SQL with regular expressions, and run it across every table

C Apply dynamic data masking to every column in every table

D Sensitive Data Protection discovery, across the organisation

◆ PAUSE AND DECIDE

What should they do?

A Manual review
Slow, and owners miss what they never knew.

B Regex SQL
Brittle, and must be maintained.

C Mask everything
Masks blindly, and finds nothing.

D SDP discovery
Profiles every table, and keeps watching.

THE SIGNAL

Find unknown personal data at scale: Sensitive Data Protection discovery.

> Westgrid wants total demand per region for each half hour, published once that half hour closes.
>
> Readings stream in from Pub/Sub, with event timestamps from the meters.

Which pipeline design fits?

A Dataflow with sliding 30-minute windows, starting every minute

B Dataflow with session windows by region, and a 30-minute gap

C Dataflow with fixed 30-minute windows, keyed by region

D A BigQuery scheduled query on the raw table, every 30 minutes

◆ PAUSE AND DECIDE

Which pipeline design fits?

A Sliding windows
Thirty overlapping totals per half hour.

B Session windows
Sessions end on gaps, not on the clock.

C Fixed windows
One total per region per half hour.

D Scheduled query
Clock time, not event time.

THE SIGNAL

Per half hour, back to back, by event time: fixed windows.

A new field engineer app stores job notes for site visits. Engineers often work in basements with no signal.

Their changes must be saved on the phone, and sync when they reconnect.

Which database fits best?

A Firestore, using its client SDKs with offline persistence

B Cloud SQL for PostgreSQL, called through the app's REST API

C Bigtable, with one row for each job note

D BigQuery, using the Storage Write API from each phone

◆ PAUSE AND DECIDE

Which database fits best?

A Firestore
Offline writes, and sync on reconnect, built in.

B Cloud SQL
Every write needs a connection.

C Bigtable
No offline sync, and built for huge scale.

D BigQuery
A warehouse, not an app database.

THE SIGNAL

Mobile app, offline, sync later: Firestore.

> A Managed Airflow DAG waits up to six hours each night for a supplier file to land in Cloud Storage.
>
> The waiting sensors occupy worker slots, and other tasks are delayed.

What should the team change?

A. Add more workers, so waiting sensors no longer block other tasks

B. Make the sensor deferrable, with a triggerer running

C. Replace the sensor with a six-hour sleep in a Python task

D. Move each task into its own Cloud Scheduler job

◆ PAUSE AND DECIDE

What should the team change?

A More workers

Pays for idle workers to wait.

B Deferrable sensor

The wait moves to the triggerer, freeing slots.

C Sleep

Still holds a slot, for longer.

D Cloud Scheduler

Loses dependencies and retries.

THE SIGNAL

Long waits in Airflow: deferrable operators and a triggerer.

Paper 1 answers

QUESTIONS	ANSWERS
1 to 5	B, D, A, C, A
6 to 10	B, D, C, D, A
11 to 15	C, B, C, D, B
16 to 20	A, D, C, A, B

REMEMBER

Check each miss against its worked answer before you move on.

Where your marks came from

SECTION	QUESTIONS	REVISIT
1 Design, 22%	2, 6, 10, 14, 17	Modules 02 to 09
2 Ingest and process, 25%	1, 7, 9, 15, 18	Modules 10 to 18
3 Store, 20%	3, 8, 13, 19	Modules 19 to 26
4 Analyse, 15%	4, 11, 16	Modules 27 to 32
5 Maintain and automate, 18%	5, 12, 20	Modules 33 to 39

THE PATTERN

One weak section costs more than a scattered few. Fix the section, not the single question.

What your total suggests

SCORE	WHAT IT SUGGESTS
17 to 20	Strong. Keep sharp with papers 2 and 3.
14 to 16	Close. Revisit the weakest section, then take paper 2.
Under 14	Rework the weak sections first. Retake this paper in a week.

HONEST LIMIT

Google does not publish a pass mark. These bands guide study. They do not predict a result.

Module 43 in five lines

- Late data that must count: **allowed lateness** and late triggers.
- Stored only in one country: **dataset location** plus the resource locations policy.
- Order per meter: an **ordering key**. Changes into BigQuery: **Datastream**.
- Data falls in value with age: **lifecycle rules**, and **partition expiration**.
- Thin link, hundreds of terabytes: do the **arithmetic**, then ship it.

COMING UP

Next: module 44, practice paper 2. Twenty more questions, some pairing with today's.